



RS6xxx Key BOM Selection Guide

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Revision History

Version	Version Description	Date
1.4	Update section 2.1 "Precautions", including precautions for selecting Flash models not listed in the support list	2026-01-19
1.3	Update section 2.1 "Note": The no-development version supports without the need to add Flash ID	2026-01-07
1.2	Add notes for using Winbond Flash	2025-09-16
1.1	Add Flash capacity selection instructions	2025-05-10
1.0	Initial release	2025-02-13

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1 Overview

1.1 Document Purpose

When using RS6xxx, it is necessary to pair it with several component devices. Some of these devices (as detailed below) have a crucial impact on the system's functional performance. Therefore, our company provides this document to assist users in quickly completing device selection.

We recommend that customers prioritize using the models marked as "Verified" In the support list below.

If the models listed are not suitable, users can select their own models based on the guidance provided in the following chapters and conduct thorough verification and testing.

This table will be continuously updated, and users can obtain the latest version through the dedicated profile publishing channel. Generally, list updates will be accompanied by SDK updates. Please consult the FAE for specific details.

1.2 Support Status Identification Description

Table 1-1 Identification Instructions

Identification	Description
M/A	This platform has passed verification and has a mass production case.
T/A	This platform has passed verification, but it is uncertain whether there are mass production cases.
L1/A	This model has not been verified on this platform, but other packages from the same series have passed verification on this platform and are theoretically supported. If you want to use it, please complete the verification yourself.
L2/A	After passing verification on other existing chip platforms, the hardware and software of this platform inherit past designs and are theoretically supported, but have not undergone practical verification. If you want to use it, please complete the verification yourself.
P/A	Based on the parameters listed in the device specification, it is assessed as supportable, but it has not been physically verified. If it is to be used, the user should complete the verification themselves.
N/A	This platform currently does not support it.

2 SPI NOR FLASH

2.1 Precautions

- When designing a scheme, users can refer to this table to determine the SPI NOR FLASH selection based on required parameters such as package, voltage range, capacity, etc.
- For the solution involving off-the-shelf firmware, if OTA functionality is required, choose a capacity of 8Mbit; otherwise, select 4Mbit. For the solution involving custom-developed firmware, choose the capacity based on the actual firmware size.
- Users are advised to choose models listed in the table as much as possible. If the desired model cannot be found in the table, for users who do not require development, they can directly use it and complete the verification themselves. For SDK development users, please refer to the relevant SDK manual instructions and complete the verification yourself.
- When users choose Flash, if they select a model already listed in the table below, they can directly verify and use it; if they select a model not listed in the table below, the default compatible parameters of the host computer are: 2-wire, support for sleep wake-up, a typical wake-up delay value of 30us, and a typical sleep delay value of 2us. Users need to confirm that the selected model meets the default compatible parameters and complete the verification on their own.
- When users verify models that are not listed in this table and encounter failures such as erasure or programming failures, they can first add a Flash ID for verification. If verification is successful after adding the Flash ID, it indicates that the default compatibility parameters do not match this model; if verification fails after adding the Flash ID, it is necessary to replace with another model or check whether there is a problem with the hardware circuit.

2.2 Support List

Table 2-1 SPI NOR FLASH Support List

Manufacturer	Model	Capacity (Mbit)	Voltage (V)	Encapsulation	IO Pattern	MAX Frequency (MHz)	Support State	Remark
GD	GD25Q40ETIG	4	2.7~3.6	SOP8-150	4	133	T/A	—
GD	GD25Q80ETIG	8	2.7~3.6	SOP8-150	4	133	T/A	—
GD	GD25WQ40ETIG	4	1.7~3.6	SOP8-150	4	104	T/A	—
GD	GD25WQ40EEIGR	4	1.7~3.6	USON8	4	104	T/A	—
GD	GD25WQ80ETIG	8	1.7~3.6	SOP8-150	4	104	T/A	—
GD	GD25WQ80EEIGR	8	1.7~3.6	USON8	4	104	T/A	—
XMC	XM25QH40BJ	4	2.7~3.6	SOP8-150	4	104	L2/A	—
XMC	XM25QH40BU	4	2.7~3.6	USON8	4	104	T/A	—
XMC	XM25QH80BJ	8	2.7~3.6	SOP8-150	4	104	T/A	—
XMC	XM25QH80BU	8	2.7~3.6	USON8	4	104	T/A	—
WINBOND	W25X40CLSING	4	2.3~3.6	SOP8-150	2	104	T/A	MOSI/MISO needs to be pulled up to VCC through a 100K resistor to avoid leakage during sleep mode.

Manufacturer	Model	Capacity (Mbit)	Voltage (V)	Encapsulation	IO Pattern	MAX Frequency (MHz)	Support State	Remark
WINBOND	W25Q80DVSNI	8	2.7~3.6	SOP8-150	4	104	T/A	MOSI/MISO needs to be pulled up to VCC through a 100K resistor to avoid leakage during sleep mode.
PUYA	P25Q40SH-SSH-IR	4	2.3~3.6	SOP8-150	4	120	T/A	—
PUYA	P25Q80SH-SSH-IR	8	2.3~3.6	SOP8-150	4	120	T/A	—
PUYA	P25Q80SH-UXH-IR	8	2.3~3.6	USON8	4	120	T/A	—
PUYA	P25Q80H-SSH	8	2.3~3.6	SOP8-150	4	104	T/A	—
PUYA	P25Q40SU-SSH-IR	4	1.7~3.6	SOP8-150	4	85	T/A	—
PUYA	P25Q80SU-SSH-IR	8	1.7~3.6	SOP8-150	4	85	T/A	—
XTX	XT25F08FDTIGT	8	2.3~3.6	USON8	4	108	T/A	—
ONWAY	ZB25VQ40BDJG	4	2.3~3.6	USON8	4	133	T/A	—

3 DCDC

3.1 Precautions

- When designing a scheme, if the user determines the need to use a DCDC, they can select a model through this table.
- For models marked with "**P/A**", it indicates that they are evaluated to be usable but have not been physically verified. Users are required to conduct their own verification, and the verification method is provided below for reference.
- If the required model cannot be found in the table, users can select a model based on the parameter requirements in 3.2 and complete the verification. Refer to the following text for verification methods.

3.2 Parameter Requirements

- The rated output maximum current must be greater than 0.4A
- Support 1.3V output (mandatory)
- The efficiency is greater than 80% when outputting 20uA at 1.3V (optional)
- Typical static leakage is less than 600nA (optional)

3.3 Support List

Table 3-1 DCDC Support List

Manufacturer	Model	Input Voltage (V)	Rated Maximum Output Current (A)	Efficiency At 20uA Output At 1.3V (Estimated Based On The Specification Sheet)	Typical Static Electricity Leakage (nA)	Encapsulation	Support Status
Joulwatt	JW5712A	2.3~5.5	0.6	85%	360	WLCSP1.575x0.875-8	T/A
	JW5717	2.3~5.5	0.6	83%	360	DFN3*2-12	T/A
SGMICRO	SGM6027A	2.5~5.5	0.6	80%	580	WLCSP-0.8x1.6-8B	T/A
	SGM6027B	2.5~5.5	0.6	80%	580	WLCSP-0.8x1.6-8B	P/A
TI	TPS62841DLC	1.8~6.5	0.75	85%	60	SON-8	P/A
	TPS62841YBG	1.8~6.5	0.75	85%	60	WCSP-6	P/A
	TPS62841DGR	1.8~6.5	0.75	85%	60	HVSSOP-8	P/A
	TPS627431	2.15~5.5	0.4	85%	360	DSBGA (identical to JW5712A)	P/A
	TPS628437DRL	1.8~5.5	0.6	85%	275	SOT563	P/A
RICHTEK	RT5707A	2.2~5.5	0.4	83%	360	WL-CSP-8B 0.9x1.6 (BSC)	P/A
	RT5713U	2.2~5.5	0.4	80%	360	WL-CSP-6B 1.415x0.885 (BSC)	P/A
	RT5713U	2.2~5.5	0.6	80%	360	WL-CSP-6B 1.415x0.885 (BSC)	P/A

3.4 Verification Method

For specific details, please refer to the PCB performance analysis in "the RS6130 Hardware Design and PCB Layout Instruction Manual".

4 PSRAM

4.1 Precautions

- When designing a scheme, if the user determines the need to use PSRAM, they can select a model through this table. Only the RS6240 platform may use PSRAM.
- Please try to choose the models listed in the table. If you cannot find the desired model in the table, please be sure to submit a verification request through the FAE channel. Our company will add the model to the table after verification is approved.

4.2 Support List

Table 4-1 PSRAM Support List

Manufacturer	Model	Capacity (Mbit)	Voltage (V)	Encapsulation	Io Pattern	Max Frequency (MHz)	Support State	Remark
Apmemory	APS1604M-3SQR-ZR	2	2.7~3.6	USON-8	SPI/QPI@SD R mode	133	T/A	—

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